

Experiment No. 2												
SE (CSE)		ROLL NO :										
Date of Implementation:												
Aim: Implement Data Visualization techniques in Python												
Programming Language Used : Python												
Upon completion of this experiment, students will be able to CO2 : Apply various analysis and visualization techniques using modern tools on real-world data												
Indicator	Poor	Average	Good									
Timeline Maintains submission deadline (2)	Submission not done (0)	One or More than One week late (1)	Maintains deadline (2)									
Completion and Organization (2)	Many mistakes in documents (0)	Document is just acceptable (1)	Completed whole document and neatly organized (2)									
Program Performance (2)	Could not perform at all (0)	Implemented few parts (1)	Full implementation (2)									
Knowledge In depth knowledge of the Experiment (4)	Unable to answer questions(0)	Unable to answer few questions (1-2)	Able to answer all questions with proper explanation (3-4)									
Assessment Marks : <table border="1"> <tr> <td>Timeline</td> <td></td> </tr> <tr> <td>Completion and Organization</td> <td></td> </tr> <tr> <td>Program Performance</td> <td></td> </tr> <tr> <td>Knowledge</td> <td></td> </tr> </table>					Timeline		Completion and Organization		Program Performance		Knowledge	
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TOTAL :												

EXPERIMENT	2
Aim	Implement Data Visualization techniques in Python
Theory	Principles of Effective Data Visualization Effective data visualization aims to communicate information clearly, accurately, and efficiently. The key principles include: Clarity: Visuals should be simple and easy to understand without ambiguity. Accuracy: Data must be represented truthfully without distortion or misleading scales. Relevance: The visualization should match the nature of the data and the objective of analysis. Consistency: Uniform colors, scales, and labels improve readability. Simplicity: Avoid unnecessary decorations (chartjunk) that distract from insights. Comparability: Enable easy comparison across categories or time periods. Benefits of Data Visualization Data visualization provides several advantages: - Simplifies complex datasets into understandable visuals - Helps in quick identification of patterns, trends, and outliers - Improves decision-making through visual insights - Enhances communication of results to technical and non-technical audiences - Supports exploratory data analysis (EDA) and hypothesis generation Types of Visualizations Charts are commonly used to represent categorical or time-based data. - Bar Charts: Compare values across categories - Line Charts: Show trends over time - Pie Charts: Represent proportional data or percentages Graphs are mainly used for distribution analysis and relationship exploration. - Scatter Plots: Show relationships between two numerical variables - Histograms: Display frequency distribution of numerical data - Box Plots: Summarize data using quartiles and detect outliers - Word Clouds: Visualize text data based on word frequency Maps are used when data has a geographical component. - Geographic Maps: Display data across locations (countries, states, cities) - Heat Maps: Represent data intensity or correlation using color gradients Dashboards are interactive visual platforms that combine multiple visualizations. - Provide real-time insights - Allow filtering, drilling down, and comparison - Widely used in business intelligence and analytics

	Data Visualization Tools Several tools are available for creating effective visualizations: Microsoft Excel: Easy-to-use tool for basic charts and dashboards Tableau: Advanced visualization tool with drag-and-drop features Power BI: Business intelligence tool for interactive reports and dashboards
Implementation	Using Excel : 1. Load the Dataset. Download Students Performance in Exams dataset from Kaggle 2. Open the CSV file in Microsoft Excel 3. Calculate average Math, Reading, Writing scores and show in bar chart 4. Count number of Male and Female students and show in pie chart 5. Select first 10 student records. Show the score trend by line chart Using Python : 1. Load any data set that is available in Python such as IRIS data set, TITANIC data set or TIPS data set or any other data set available. 2. Show bar chart (average feature values), stacked bar chart, line graph, Histogram, scatter plot, box plot, parallel coordinate plot, heatmap, QQ plot, violin plot, pair plot, KDE plot, bubble chart
Conclusion	Comment on the different graphs output.
Post Lab Questions:	1. What is KDE plot? 2. What is violin plot? 3. What is bubble chart?